

EBN 111 Practical 1: Instructions

ebn2017pracs@gmail.com

Introduction

The goal of this practical is to familiarize yourself with OrCAD Capture and PSPICE, which will be used to simulate DC circuits. Practical 1 will consist of three parts:

1. A pre-practical assignment that must be completed individually in the week that precedes practical 1.
2. A pre-practical test that must be completed at the start of the practical session.
3. The practical assignment itself that will be completed in groups of two during the allocated practical session.

Tasks II and III will take place during your allocated practical session. You can view your allocated session at *clickUp* → *EBN 111* → *Practicals/Prakties*. The venue for all practicals is the CAEC labs in the ENG2 building.

Task I

In order to complete the assignment, you will need to use the OrCAD software suite. A tutorial has been created that you can use to both install the software and familiarize yourself with the program's functionality.

The tutorial is available at: *clickUp* → *EBN 111* → *Practicals/Prakties* → *Practical 1* → *Practical 1 Tutorial*.

After completion of the tutorial you can start the pre-practical assignment. The assignment is available at: *clickUp* → *EBN 111* → *Practicals/Prakties* → *Practical 1* → *Practical 1 pre-prac assignment*.

You must bring the completed pre-practical assignment to the practical session. The assignment does not count towards your marks but you will not be allowed to enter the session if it is not completed.

Task II

The pre-practical test will be written on clickUP at the start of your practical session and the questions will test you in the following areas:

- Information provided during the pre-practical lecture.
- Information contained in the slides provided for practical 1.
- Information obtained from completing the tutorial.
- Information obtained from completing the pre-practical assignment itself.

The test will take up to 30 minutes of the practical session. Note that this test is closed-book. Hence you will be required to complete it without reference to any of the study material. After completing the test, you will move on to the practical itself.

Task III

The practical itself must be completed during the practical session. Students may work in groups of 2.

Please remember a pen and paper to write down your answers to the practical questions. The Practical 1 Guidelines will be available on clickUP during the practical. Follow the instructions, perform the simulations and then write down your answers. After you have completed the practical and have filled in all the values, go onto clickUP and complete the Practical 1 Evaluation.

You will not submit your hard copy of the answers, but the answers which you have noted here must be submitted to the Practical 1 Evaluation on clickUP. This evaluation will only be open during the time of your practical session, therefore must be completed within these three hours.

Conclusion

In order to pass the practical, you have to complete and bring the pre-practical assignment to the practical, and also submit the following clickUP evaluations:

1. Pre-practical Test
2. Practical 1 Online Evaluation Please

remember:

1. You will not be allowed in the labs if are wearing open shoes. You can familiarize yourself with the lab rules at <http://www.up.ac.za/en/eece/article/2008415/undergraduate-laboratory-rules>

Marks will be released at the end of the week, once all groups have completed their practical sessions.